

Engineering Sprint 8 – Designing Asynchronous Workflows

Objective

The objective of Engineering Sprint 8 is to understand how to design reliable asynchronous workflows in Salesforce. The sprint teaches how to separate immediate user operations from background processing using Future Methods, Queueable Apex, Batch Apex, and Scheduled Apex. It also explains monitoring asynchronous jobs and handling failures effectively.

Introduction

Asynchronous Apex allows long-running operations to execute in the background without making users wait. Before selecting an Apex mechanism, developers should first decide which tasks must complete immediately and which can safely execute later. This helps improve system performance, scalability, and user experience.

1. Queueable Apex

Definition

Queueable Apex is used to execute background processing after the main transaction completes. It is suitable for structured asynchronous work where users should not wait for secondary operations.

Code

```
public class OfferPostProcessingJob implements Queueable {  
    private Id offerId;  
  
    public OfferPostProcessingJob(Id offerId){  
        this.offerId = offerId;  
    }  
  
    public void execute(QueueableContext context){  
        System.debug('Job A');  
        System.enqueueJob(new NotificationJob());  
    }  
}
```

Execute Anonymous

```
Id sampleId='0010000000000001AAA';  
  
System.enqueueJob(new OfferPostProcessingJob(sampleId));
```

Output

The Queueable job executes in the background and starts another Queueable job.

The screenshot shows the Salesforce Developer Console interface. At the top, the browser address bar displays the URL: `curious-raccoon-5f1zx8-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage`. Below the browser window, the 'Execution Log' tab is active, showing a table of log entries:

Timestamp	Event	Details
10:41:43:009	USER_DEBUG	[11]DEBUG[Queueable Job Started
10:41:43:009	USER_DEBUG	[13]DEBUG[Offer ID : 0010000000000001AAA
10:41:43:009	USER_DEBUG	[15]DEBUG[Background Processing Completed

Below the Execution Log, the 'Logs' tab is active, showing a table of log entries:

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA SETHA PRAVALLIKA	Unknown	QueueableHandler	8/7/2026, 10:41:43 AM	Success		3.26 KB
PADALA DURGA SETHA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeAn...	8/7/2026, 10:41:42 AM	Success	Unread	3.7 KB

2. Future Method

Definition

Future Methods are used for asynchronous processing in existing Salesforce applications. The sprint explains that working Future Methods should not be replaced unless there is a valid technical or business reason.

Code

```
public class FutureExample {  
    @future  
    public static void sendNotification(){  
        System.debug('Future Method Running');  
    }  
}
```

Execute Anonymous

```
FutureExample.sendNotification();
```

The screenshot shows the Salesforce Developer Console interface. The top bar indicates the user is logged in as 'curious-raccoon-5flzx8-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage'. The 'Execution Log' tab is active, displaying a table with columns: Timestamp, Event, and Details. The log shows a single entry at 10:44:31.031, event 'USER_DEBUG', with details '[7][DEBUG]Future Method Running'.

Below the Execution Log, the 'Logs' tab is active, showing a table with columns: User, Application, Operation, Time, Status, Read, and Size. The table contains four entries, all with a status of 'Success'.

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA SETHA PRAVALIKA	Unknown	FutureHandler	8/7/2026, 10:44:31 AM	Success	Read	2.83 KB
PADALA DURGA SETHA PRAVALIKA	Unknown	/services/data/v67.0/tooling/executeIn...	8/7/2026, 10:44:30 AM	Success	Unread	2.05 KB
PADALA DURGA SETHA PRAVALIKA	Unknown	QueueableHandler	8/7/2026, 10:41:43 AM	Success	Unread	3.26 KB
PADALA DURGA SETHA PRAVALIKA	Unknown	/services/data/v67.0/tooling/executeIn...	8/7/2026, 10:41:42 AM	Success	Unread	3.7 KB

The bottom of the console shows a Windows taskbar with the date 07-08-2026 and time 10:45.

3. Queueable Chaining

Definition

Queueable Chaining allows one Queueable job to start another Queueable job after successful execution. This keeps responsibilities separate and improves maintainability.

Code

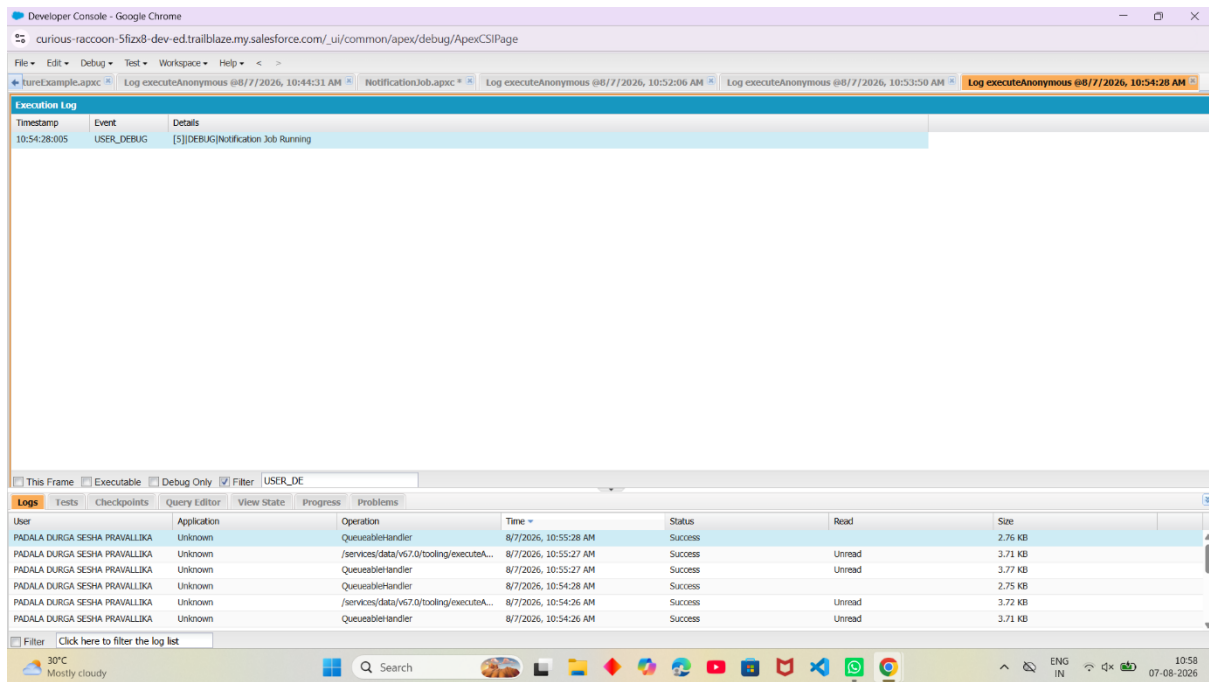
```
public class NotificationJob implements Queueable{

    public void execute(QueueableContext context){

        System.debug('Notification Job Running');

    }

}
```



4. Batch Apex

Definition

Batch Apex processes very large datasets efficiently by dividing records into smaller batches. It uses three methods: `start()`, `execute()`, and `finish()`.

Code

```
public class StudentBatch implements Database.Batchable<SObject>{

    public Database.QueryLocator start(Database.BatchableContext bc){

        return Database.getQueryLocator(

            'SELECT Id,Name FROM Account'

        );

    }

    public void execute(Database.BatchableContext bc,List<Account> scope){

        for(Account a:scope){

            System.debug(a.Name);

        }

    }

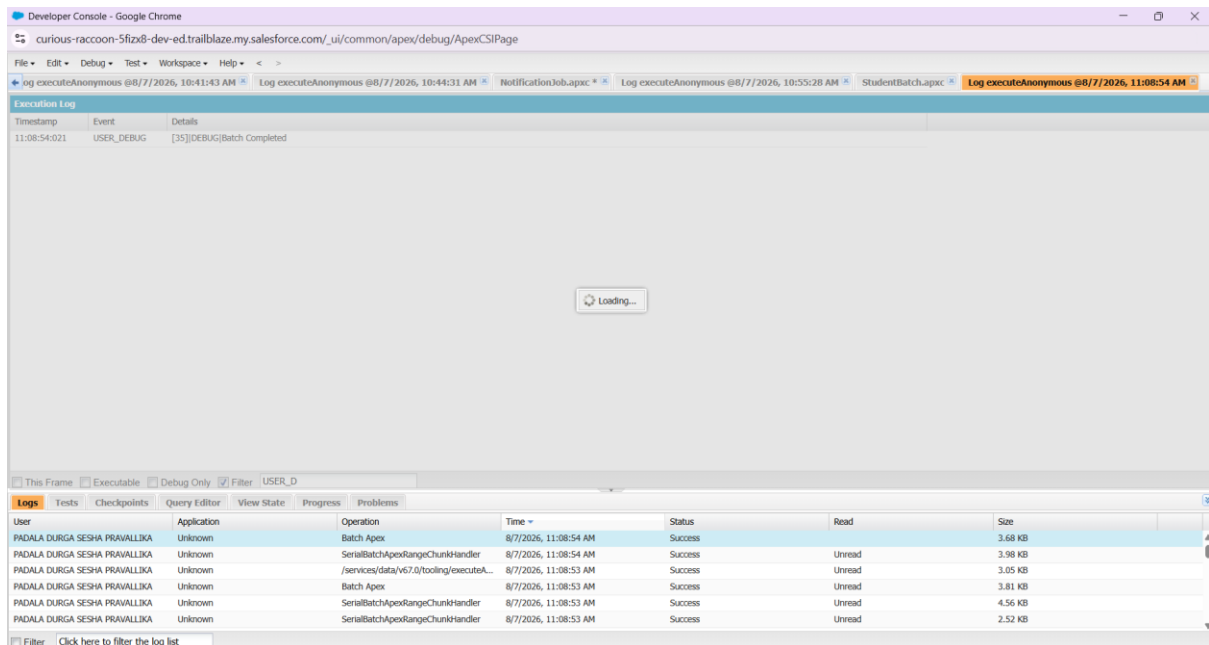
    public void finish(Database.BatchableContext bc){

        System.debug('Batch Completed');
```

```
}
}
```

Execute Anonymous

```
Database.executeBatch(new StudentBatch(),5);
```



5. Scheduled Apex

Definition

Scheduled Apex executes Apex automatically at a specified time using a CRON expression. It is useful for recurring business activities.

Code

```
public class DailyScheduler implements Schedulable{
    public void execute(SchedulableContext sc){
        Database.executeBatch(new StudentBatch(),5);
    }
}
```

Execute Anonymous

```
String cron='0 0 12 * * ?';
```

```
System.schedule(
```

```
'Batch Scheduler',
```

```
cron,
```

```
new DailyScheduler()
```

);

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'sched' and a list of navigation items: Feature Settings, Sales, Products, Product Schedules Settings, Scheduled Reminders, Scheduled Reminder Settings, Service, Field Service, Scheduling Console Settings, Environments, and Jobs. The 'Scheduled Jobs' item is highlighted. The main content area is titled 'Scheduled Jobs' and includes a 'Help for this Page' link. Below the title, there is a message: 'The All Scheduled Jobs page lists all of the jobs scheduled by your users. Multiple job types may display on this page. You can delete scheduled jobs if you have the permission to do so.' A yellow banner indicates 'Percentage of Scheduled Jobs Used: 1%' and provides a link to 'Lightning Platform Apex Limits'. Below this, there is a table of scheduled jobs. The table has columns: Action, Job Name, Submitted By, Submitted, Started, Next Scheduled Run, Type, and Cron Trigger ID. The table lists two jobs: 'Daily Job' and 'Metalytics Data Loader Job for Org : 000hg000002m97'. The bottom of the screen shows a Windows taskbar with the date 07-08-2026 and time 11:19.

Action	Job Name	Submitted By	Submitted	Started	Next Scheduled Run	Type	Cron Trigger ID
Manage Del Pause Job	Daily Job	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:10 am		08/08/2026, 10:00 am	Scheduled Apex	08ehg000000qfYY
Del	Metalytics Data Loader Job for Org : 000hg000002m97	User Integration	26/07/2026, 10:08 am	07/08/2026, 5:07 am	08/08/2026, 5:07 am	Autonomous Data Loader Job	08ehg000000cLTD

6. Apex Jobs Monitoring

Definition

Salesforce provides the **Apex Jobs** page to monitor asynchronous operations. Developers can verify whether jobs are queued, running, completed, or failed and investigate issues if necessary.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'apex' and a list of navigation items: Email, Apex Exception Email, Custom Code, Apex Classes, Apex Settings, Apex Triggers, Application Test Execution, Application Test History, Environments, and Jobs. The 'Apex Jobs' item is highlighted. The main content area is titled 'Apex Jobs' and includes a 'Help for this Page' link. Below the title, there is a message: 'The Apex Jobs page lists all of the jobs scheduled by your users. Multiple job types may display on this page. You can delete scheduled jobs if you have the permission to do so.' A yellow banner indicates 'Percentage of Scheduled Jobs Used: 1%' and provides a link to 'Lightning Platform Apex Limits'. Below this, there is a table of scheduled jobs. The table has columns: Action, Submitted Date, Job Type, Status, Status Detail, Total Batches, Batches Processed, Failures, Submitted By, Completion Date, Apex Class, Apex Method, and Apex Job ID. The table lists 12 jobs, including 'Daily Scheduler', 'StudentBatch', 'NotificationJob', and 'OfferPostProcessingJob'. The bottom of the screen shows a Windows taskbar with the date 07-08-2026 and time 11:31.

Action	Submitted Date	Job Type	Status	Status Detail	Total Batches	Batches Processed	Failures	Submitted By	Completion Date	Apex Class	Apex Method	Apex Job ID
	07/08/2026, 11:24 am	Scheduled Apex	Queued		0	0	0	DURGA SETHA PRAVALLIKA PADALA		DailyScheduler		707hg000001jKJ4
	07/08/2026, 11:10 am	Scheduled Apex	Aborted		0	0	0	DURGA SETHA PRAVALLIKA PADALA		DailyScheduler		707hg000001jKJ17
	07/08/2026, 11:08 am	Batch Apex	Completed		3	3	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:08 am	StudentBatch		707hg000001jKJUN
	07/08/2026, 11:07 am	Queueable	Completed		0	0	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:07 am	NotificationJob		707hg000001jPo6
	07/08/2026, 11:07 am	Queueable	Completed		0	0	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:07 am	OfferPostProcessingJob		707hg000001jKJc
	07/08/2026, 11:05 am	Queueable	Completed		0	0	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:05 am	NotificationJob		707hg000001jBgw
	07/08/2026, 11:04 am	Queueable	Completed		0	0	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:04 am	NotificationJob		707hg000001jBwt
	07/08/2026, 11:04 am	Queueable	Completed		0	0	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:04 am	OfferPostProcessingJob		707hg000001jFoO
	07/08/2026, 11:01 am	Queueable	Completed		0	0	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:01 am	NotificationJob		707hg000001jK7J
	07/08/2026, 11:01 am	Queueable	Completed		0	0	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:01 am	OfferPostProcessingJob		707hg000001jL3B
	07/08/2026, 11:00 am	Batch Apex	Completed		3	3	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:00 am	StudentBatch		707hg000001jD9K
	07/08/2026, 11:00 am	Batch Apex	Completed		3	3	0	DURGA SETHA PRAVALLIKA PADALA	07/08/2026, 11:00 am	StudentBatch		707hg000001jKot